

**[BUG] `qml.pow()` is not working properly with the interfaces**

<https://github.com/PennyLaneAI/pennylane/issues/4825>

ROW 116

# [BUG] `qml.pow()` is not working properly with the interfaces #4825

[New issue](#)Closed

#4827



KetpuntoG opened on Nov 10, 2023 - edited by KetpuntoG

Edits ...

## Expected behavior

I would expect this code work

```
import tensorflow as tf

dev = qml.device("default.qubit")
@qml.qnode(dev)
def circuit(theta):
    qml.pow(qml.RX(theta, wires = 0), z=3)
    return qml.expval(qml.PauliZ(0))

phi = tf.Variable(1.0)
circuit(phi)
```

If we change and set `z = 3.0`, the code will work.  
However, in `torch` the behavior is the opposite. This code will not work:

```
import torch

dev = qml.device("default.qubit")
@qml.qnode(dev)
def circuit(theta):
    qml.pow(qml.RX(theta, wires = 0), z=3.)
    qml.ctrl(qml.pow(qml.RX(theta, wires = 0), z=3), control = 3)
    return qml.expval(qml.PauliZ(0))

phi = torch.tensor(1.0, requires_grad = True)
circuit(phi)
```

## Actual behavior

```
KeyError                                Traceback (most recent call last)
[/usr/local/lib/python3.10/dist-packages/autoray/autoray.py](https://8x3darvacxn-496ff2e9c6d22116-0-cola...-0f
  472     try:
--> 473         lib_fn = _FUNCS[backend, fn]
      474     except KeyError:

KeyError: ('tensorflow', 'linalg.matrix_power')
```

## Additional information

Christina has identified that if you put `lazy = False` the code works. Also it seems to be related with the type of `z`  
I think Jax is working like Torch

## System information

```
Name: PennyLane
Version: 0.34.0.dev0
Summary: PennyLane is a Python quantum machine learning library by Xanadu Inc.
Home-page: https://github.com/PennyLaneAI/pennylane
Author:
Author-email:
License: Apache License 2.0
Location: /usr/local/lib/python3.10/dist-packages
Requires: appdirs, autograd, autoray, cachetools, networkx, numpy, pennylane-lightning, requests, rustworkx, s
Required-by: PennyLane-Lightning

Platform info:          Linux-5.15.120+-x86_64-with-glibc2.35
Python version:         3.10.12
Numpy version:          1.23.5
Scipy version:          1.11.3
Installed devices:
- lightning.qubit (PennyLane-Lightning-0.33.1)
- default.gaussian (PennyLane-0.34.0.dev0)
```

### Assignees

No one assigned

### Labels

bug

### Type

No type

### Projects

No projects

### Milestone

No milestone

### Relationships

None yet

### Development

fallback to scipy when linalg.matrix\_power isn't available  
PennyLaneAI/pennylane

### Notifications

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### Participants



```
default.qubit.tf (PennyLane-0.34.0.dev0)
- default.qubit.torch (PennyLane-0.34.0.dev0)
- default.qutrit (PennyLane-0.34.0.dev0)
- null.qubit (PennyLane-0.34.0.dev0)
```

### Existing GitHub issues

- ☒ I have searched existing GitHub issues to make sure the issue does not already exist.



KetpuntoG added **bug** on Nov 10, 2023

KetpuntoG changed the title **[BUG] `qml.pow()` is not working properly with `tensorflow`** **[BUG] `qml.pow()` is not working properly with the interfaces** on Nov 10, 2023

KetpuntoG mentioned this on Nov 10, 2023

[Adding functionality `iterative\_gpe` #4804](#)



timmysilv on Nov 10, 2023

Contributor ...

the torch code is working for me locally - can you confirm your torch version is up-to-date?

as for tensorflow, it seems like it doesn't have `matrix_power` implemented, and giving a float makes us use `scipy.linalg.fractional_matrix_power`. perhaps we should also use that if our data is tensorflow data. I'll open a PR



timmysilv mentioned this on Nov 10, 2023

[fallback to scipy when `linalg.matrix\_power` isn't available #4827](#)



KetpuntoG on Nov 13, 2023 · edited by KetpuntoG

Edits Contributor Author ...

Yes it works, I forgot to put the point on the second `z = 3`.  
If you add it, it will not work



timmysilv on Nov 13, 2023

Contributor ...

ah yeah, i'm seeing it with a floating-point `z` and a torch tensor with `requires_grad=True`. I'm not sure if there's anything that can be done short-term in that case



timmysilv closed this as **completed** in [#4827](#) on Nov 16, 2023

timmysilv added a commit that references this issue on Nov 16, 2023

[fallback to scipy when `linalg.matrix\_power` isn't available \(#4827\)](#) ...

Verified 8edac6f

# fallback to scipy when linalg.matrix\_power isn't available #4827

<> Code

**Merged** timmysilv merged 6 commits into master from tf-matrix-power on Nov 16, 2023

Conversation 6 Commits 6 Checks 0 Files changed 6 +48 -14



timmysilv commented on Nov 10, 2023 · edited

Contributor

**Context:**

Tensorflow does not have a `matrix_power` function. For non-int powers, we always fall back to `scipy`, and we noticed that this works with Tensorflow as well! We can use this...

**Description of the Change:**

- Whenever trying to raise TF matrices to an integer matrix-power, use `scipy.linalg.fractional_matrix_power`. Until today, we just raised an error. `Pow` is not trainable regardless, so there's no harm in adding this.
- This also indirectly allowed for raising any `QubitUnitary` to a non-int power (eg: `qml.QubitUnitary(np.eye(2), wires=[0]).pow(3.5)`)

**Benefits:**

Less errors! more features!

**Possible Drawbacks:**

- Still not differentiable
- `QubitUnitary.pow` looks more complex to support more stuff

**Related GitHub Issues:**

Fixes #4825 [sc-49952]



Reviewers

lillian542



albi3ro



KetpuntoG



Assignees

No one assigned

Labels

None yet

Projects

None yet

Milestone

No milestone

Development

Successfully merging this pull request may close these issues.

**[BUG]** `qml.pow()` is not working properly wi...

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3 participants

timmysilv added 2 commits 2 years ago

fallback to scipy when linalg.matrix\_power isn't available

4c61950

fix test; don't allow batching for edge-case QubitUnitary.pow

79d483c



timmysilv commented on Nov 10, 2023

View reviewed changes